

Workshop 1: the Top500 list

1 Introduction

The Top500 list ranks the fastest supercomputers in the world according to their performance on the LINPACK benchmark. The list has run since 1993 and is updated twice a year, providing a long term view of trends in the field. In this workshop we will explore the data available on the Top500 web site and see what it tells us about trends in supercomputing.

2 The top 10 now and then

In this section you will compare a recent top 10 list with previous top 10 lists to see how the most powerful computers in the world have changed over time.

- The top 10 list in November 2025 can be found at <https://top500.org/lists/top500/2025/11/>. The number 1 system on this list is “El Capitan”. How many cores does it have and what is its LINPACK performance $R_{\max}$? What is its LINPACK efficiency ($R_{\max}/R_{\text{peak}}$)?
- The first list, published in June 1993, can be found at <https://top500.org/lists/top500/1993/06/>. The number 1 system on this list is a Thinking Machines Corporation (TMC) CM-5. How many cores did it have and what was its LINPACK performance $R_{\max}$? What was its LINPACK efficiency ($R_{\max}/R_{\text{peak}}$)?
- How do the number of cores, LINPACK performance and LINPACK efficiency compare between “El Capitan” and the fastest TMC CM-5?
- The June 1993 top 10 includes systems which used two different vector architectures: the NEC SX-3 and Cray Y-MP systems. How do the number of cores and LINPACK efficiency of these machines compare to the TMC CM-5 and “El Capitan”?
- How have the top supercomputers changed between 1993 and the present? How do the increases in core count and LINPACK performance compare? How has the LINPACK efficiency changed?

3 Processor trends

This section looks at processor trends using the “list statistics” page (<https://top500.org/statistics/list/>). On the list statistics page select “Cores per socket” and generate statistics for the current list using the submit button. This will generate two pie charts (showing a breakdown by number of systems and weighted by performance) and a table.

- What is the most common number of cores per socket on the current list?
- What are the maximum and minimum numbers of cores per socket on the current list?
- How has this changed compared to 10 years ago and 20 years ago?

We can also see how the statistics have changed over time (<https://top500.org/statistics/overtime/>). On this page request a plot showing the systems share of cores per socket. There is a slider at the top of the chart which can be used to change the range on the horizontal (time) axis.

- When did the number of cores per socket start increasing?
- Is this consistent with the trends in microprocessor design shown in figure 1?
- What implications does this have for supercomputer applications if they are to make effective use of current and future supercomputers?

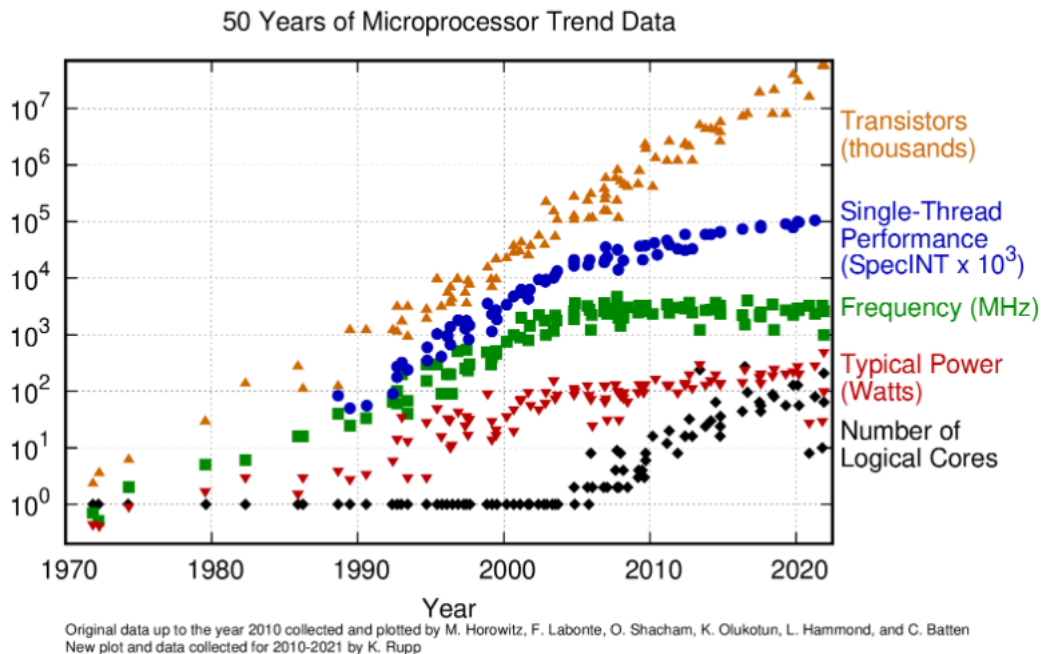


Figure 1: Trends in micro-processor design over a 40 year time span (figure from <https://github.com/karlrupp/microprocessor-trend-data>).

4 Accelerators and Co-Processors

Some supercomputers increase their performance by using an accelerator or co-processor in addition to the CPU. On the development over time page (<https://top500.org/statistics/overtime/>) request a plot of “Accelerators/CP family” to see how the use of accelerators and co-processor has changed over time. You may also want to look at individual years on the list statistics page (<https://top500.org/statistics/list/>).

- What was the first accelerator to appear on the list and when did it appear?
- What was the second accelerator to appear on the list?
- What type of accelerators are most popular now? When did they start to appear?
- What fraction of systems on the current list make use of accelerators?
- How important do you think are accelerators for HPC currently?

5 Further information

This workshop has looked at some trends in supercomputing since 1993 based on the Top500 list. If you would like to find out more about the history of supercomputing there is a presentation from the Computer History Museum: <https://www.computerhistory.org/revolution/supercomputers/10>